

The Attribution–Compression Frontier in Retrieval-Augmented Generation

Deepanshu Mody

Center for Data Science

New York University

dm6262@nyu.edu

Abstract

Context compression reduces generator input in retrieval-augmented generation, but answer quality alone does not characterize citation attribution. We measure citation attribution across compression methods and budgets, comparing reranking, extractive selection, abstractive summarization, token pruning, and an extract-cluster-rewrite construction on ASQA and QASPER under a fixed generator and primary entailment evaluator. On ASQA at a nominal 0.25 budget (achieved compression 0.08), a RECOMP-style compressor’s citations score 0.86 precision against its summaries but 0.12 against source spans under our re-attributability protocol. These estimates depend on a shared NLI model for span recovery and citation scoring and lack independent human calibration. Extractive selection’s observed grounded precision ranges from 0.43 to 0.49 across nominal budgets from one-half to one-tenth of the ASQA context, while answer quality declines. For the same RECOMP setting, claim verification after source recovery yields an unsupported rate of 0.88 versus 0.17 when checking summaries. This gap persists beyond structural rejection of missing provenance, but remains evaluator-dependent. A 200-question TRUE T5-XXL audit also finds emitted–grounded gaps under both fixed and recomputed source mappings, without establishing human-calibrated support rates.

1 Introduction

Context compression reduces generator input in retrieval-augmented generation (RAG) (Xu et al., 2024; Jiang et al., 2024; Pan et al., 2024; Chirkova et al., 2025). Citation-grade attribution requires *locatable* evidence: a reader must be able to retrieve and check the cited source (Rashkin et al., 2023; Bohnet et al., 2022). Abstractive compression can interrupt this link when citations identify compressor-generated text without its supporting source spans. We call it *attribution laundering*

when consistency with compressor-generated text is treated as evidence of support from original sources. On ASQA at nominal budget 0.25 (achieved ratio 0.08), a RECOMP-style compressor’s citations score 0.86 precision against its summaries and 0.12 against source spans under our recovery protocol. These automatic scores depend on a shared NLI evaluator and bounded candidate search; they have not been independently calibrated.

Related work already identifies a divergence between answer quality and citation grounding under compression (Li et al., 2026a), while citation support does not establish causal reliance on evidence (Wallat et al., 2025). We examine this problem across compressor families and two datasets, reporting achieved context fractions, explicit source-recovery rules, and paired comparisons on shared questions. Our measurements concern source support under an automatic evaluation protocol; they do not test causal citation faithfulness.

We conduct a controlled measurement study with a fixed generator (Qwen2.5-7B-Instruct with ALCE-style numbered-context citation prompting), a fixed primary NLI evaluator, and nominal budgets of $\{0.5, 0.25, 0.1, 0.05\}$ in generator-tokenizer tokens. Our contributions are:

- **The attribution–compression frontier.** A comparison across reranking, extractive selection, abstraction, token pruning, ECR, and an identity reference against achieved context fractions. All settings share 944 successful ASQA questions and 976 QASPER questions from 271 papers; paired tests quantify their differences.
- **A re-attributability protocol.** A unit without declared provenance is counted as re-attributable when NLI predicts entailment from a candidate source snippet; recovered sources support *grounded* citation scores

alongside *emitted* scores against compressed text.

- **Three verification modes.** We separate consistency with summaries, availability of declared provenance, and claim support after source recovery. Structural rejection under declared-only checking is distinguished from measured rejection after recovery.
- **An extract-cluster-rewrite (ECR) observation.** A simple construction greedily merges near-duplicate sentences and cites *all* members of each cluster; paired comparisons quantify its measured differences from extraction across budgets.

2 Related Work

Context compression for RAG. Text compressors select or rewrite retrieved material. Selection approaches include query-aware token pruning (Jiang et al., 2024), distilled token classification (Pan et al., 2024), sentence pruning with reranking (Chirkova et al., 2025), and evidentiality-guided selection (Jeong et al., 2025). RECOMP trains extractive and abstractive compressors for end-task performance (Xu et al., 2024), while FaviComp combines retrieved context with parametric knowledge during rewriting (Jung et al., 2025). Whether their outputs retain locatable source links depends on the integration: our extractive and ECR implementations declare source IDs, while our LLMingua-2 integration omits a token-to-source mapping. Soft compressors use learned representations, including gist tokens (Mu et al., 2023), ICAE memory slots (Ge et al., 2024), document embeddings (Cheng et al., 2024), and multi-context embeddings (Rau et al., 2025); these are outside our experiments.

Attributed generation and citation evaluation.

The AIS framework evaluates whether statements are supported by identified sources (Rashkin et al., 2023). Automatic attribution and entailment evaluation (Bohnet et al., 2022; Honovich et al., 2022) inform the citation precision and recall metrics of ALCE (Gao et al., 2023), which we adapt. QASPER provides questions over research papers with annotator-marked evidence (Dasigi et al., 2021). LongCite studies fine-grained citation generation with long contexts (Zhang et al., 2025), a complementary direction to evaluating compression under a fixed generator.

Compression and citation reliability. Li et al. (2026a) study Self-RAG on ASQA with LLMingua-2 and prefix truncation, finding sharper degradation in citation grounding than answer quality, and investigate hierarchical evidence retention. Our study adds cross-family comparisons on ASQA and QASPER, distinguishes declared and recovered source mappings, and reports emitted and grounded scores against achieved context fractions. The conceptual finding overlaps; our contribution is experimental coverage and explicit evaluation and reporting procedures, rather than priority for the divergence. Wallat et al. (2025) distinguish citation correctness from causal faithfulness and investigate post-rationalization. Our NLI-based support scores do not establish whether the generator causally relied on the cited evidence. ALCE’s summary-context ablation also illustrates a correctness–citation trade-off (Gao et al., 2023); information-preservation studies examine grounding under compression (Łajewska et al., 2025), and human evaluations examine verifiability as output abstractiveness changes (Worledge et al., 2024).

Provenance and structured representations.

Related provenance-oriented designs include claim-anchored multi-document summarization in CAMS (Guan, 2026), budgeted evidence retention in agent memory (Li et al., 2026b), and provenance-aware tiered memory (Zhu et al., 2026). Structured retrieval indexes use atomic propositions (Chen et al., 2024), recursive summary trees (Sarthi et al., 2024), or graph community summaries (Edge et al., 2024). These designs motivate explicit links between condensed representations and source evidence, but are not evaluated as compressors here. Attention-based explanations remain contested (Jain and Wallace, 2019; Wiegrefe and Pinter, 2019). Slot-based representations (Locatello et al., 2020; Tan and Kok, 2025; Ibtehaz and Kihara, 2026) suggest another possible compression mechanism; whether their internal structure can provide validated source attribution is an untested future direction.

3 Measuring Attribution Under Compression

3.1 Setup

Our pipeline is retrieve → compress → generate → evaluate. A compressor maps the retrieved

snippets (sentence-segmented, each with a stable source id, *sid*) to a sequence of *units*, and every unit either declares provenance—the set of sids it was selected from, as in extractive families—or declares none, as in abstractive families. The generator sees the units as a numbered context and is prompted, ALCE-style, to attach bracketed unit citations to each answer sentence (Gao et al., 2023). Citation quality is scored per answer statement with an NLI evaluator in the attributable-to-identified-sources sense (Rashkin et al., 2023; Bohnet et al., 2022): a statement is *recalled* iff the concatenation of its resolvable cited sources entails it, and a citation is *precise* iff that concatenation entails the statement and, in addition, the cited source alone entails it or removing that source breaks the concatenation’s entailment (Gao et al., 2023). We use DeBERTa-v3-base (MNLI+FEVER+ANLI) at threshold $\tau=0.5$, truncating each NLI input to 512 model tokens. Entailment decisions below are this model’s predictions, not independently validated judgments of source support. Per-example harness failures occur only in identity (4 of 948 on ASQA, 29 of 1,005 on QASPER); every primary setting excludes those same question IDs, with failure counts retained.

3.2 Emitted vs. Grounded Citations

We score every run under two views. We write precision as P and recall as R , with subscripts e for emitted and g for grounded. **Emitted** scores citations against the unit texts the generator actually saw; it measures consistency with the compressed context, and is what one implicitly measures when compressed units are treated as citable documents. **Grounded** first resolves each citation to retrieved source spans and scores against those. Units with declared provenance resolve to their declared sids. Units without declared provenance pass through our *re-attributability* protocol: rank source snippets by lexical similarity to the unit, run NLI on the top 10 candidates, and keep those judged to individually entail the unit at τ . A citation with no recovered span receives no precision credit and cannot support recall under this protocol. The *re-attribution rate* is the fraction of units without declared provenance with at least one recovered span, averaged across examples.

Evaluator dependence. The same NLI model filters recovered spans and scores citation support, creating a circularity risk: selection and evaluation

errors can be correlated. Lexical retrieval can miss paraphrases, and the single-snippet test can reject units supported only by combined evidence. Thus failed recovery does not establish that support is absent from the source, and the direction of overall scoring bias is unknown. Appendix C reports a second-evaluator sensitivity check; human calibration remains absent.

The emitted–grounded gap is an operational diagnostic of *attribution laundering*: claims receive credit against compressed text that is not supported through the source mappings accepted by our protocol. It does not by itself distinguish missing evidence from recovery or NLI errors.

Nominal vs. achieved compression. Budgets are nominal fractions $\{0.5, 0.25, 0.1, 0.05\}$ of each example’s retrieved context, enforced in generator-tokenizer tokens. Extractive families track their budgets closely, but abstractive compressors emit what their decoder emits: RECOMP realizes achieved ratios of 0.05–0.08 on ASQA and ≈ 0.015 on QASPER across all four nominal budgets—distinct nominal operating points collapse together. All curves therefore use the *achieved* ratio on the compression axis; plotting against nominal budgets would misplace abstractive systems by more than an order of magnitude in the worst case.

3.3 Three Verification Modes

We decompose each answer into claims (its statements with citation markers stripped; cf. atomic-fact decomposition, Min et al., 2023) and check each claim against evidence pooled from its cited units. **Summary-checked** (U_m) pools the cited units’ own texts. **Span-checked** (U_s) pools only the *declared* source spans of the cited units; claims without such evidence count as unsupported without an NLI call. We retain the label “span-checked” for this declared-only mode, which differs from grounded citation scoring: it does not recover missing spans. Its unsupported rate of 1.00 for methods without declared provenance is a consequence of this policy, not an empirical estimate that every claim lacks source support. **Recovered-source checking** (U_r) first resolves units to declared spans or, when absent, spans recovered by the same protocol used for grounded citation scoring. It deduplicates the pooled source IDs before verifying each claim; absent evidence still counts as unsupported. We evaluate this third mode for

RECOMP and LLMingua-2 at nominal 0.25 on both datasets. For methods whose units all declare provenance, recovered-source and declared-span checking coincide. We report generator input, logical verifier input (premise plus hypothesis), and their sum under declared-span checking as a partial token-cost proxy (Table 2).

Evidence selection on QASPER. On QASPER (Dasigi et al., 2021) we additionally evaluate the compressor’s own attribution: declared cited sids are mapped to their (whitespace-normalized) full paragraph texts and scored by set-F1 against the gold evidence paragraphs, taking the maximum over annotators per the official protocol. Runs without declared provenance have no declared-evidence F1; recovered-span scores would measure a different attribution pathway.

4 Experiments

4.1 Experimental Setup

We evaluate on two attribution benchmarks with different retrieval granularities. **ASQA** comprises 948 ambiguous factoid questions from the ALCE release (Gao et al., 2023), each paired with the top-5 GTR-retrieved passages, sentence-segmented into citable snippets; answer quality is string exact match (str-EM). **QASPER** (Dasigi et al., 2021) is the 1,005-question dev split over 281 NLP papers, using each supplied paper’s abstract and full text. Gold evidence is paragraph-level, so identity presents paragraphs as citable units while compressors operate on sentence snippets; declared sids map to paragraphs for set-F1 against gold evidence, max over annotators. The generator is Qwen2.5-7B-Instruct served via vLLM with greedy decoding and ALCE-style numbered-context citation prompting (Gao et al., 2023). Primary entailment judgments use the development-grade DeBERTa NLI evaluator of §3.1 at $\tau = 0.5$. We compare six compressors: *identity* (no compression; document-level on ASQA, paragraph-level on QASPER), cross-encoder *rerank top-k* (ms-marco-MiniLM-L-6-v2), embedding-based *extractive* sentence selection (all-MiniLM-L6-v2), the *RECOMP* abstractive summarizer (NQ checkpoint) (Xu et al., 2024), *LLMingua-2* token pruning (Pan et al., 2024), and our *extract-cluster-rewrite* (ECR) variant, which ranks sentences, greedily clusters near-duplicates by Jaccard word overlap (threshold 0.5, with a negation guard), and emits one unit per

cluster—the highest-relevance member sentence verbatim—citing *all* member sentence ids. Although LLMingua-2 prunes tokens verbatim, its output is a token stream with no unit-level sid mapping in our pipeline, so it declares no provenance and is scored with the no-provenance family alongside RECOMP. Each compressor runs at nominal budgets $\{0.5, 0.25, 0.1, 0.05\}$ of the retrieved context, enforced in generator-tokenizer tokens. Because compressors do not reliably hit their nominal budgets, the frontier figures and primary summary tables report the *achieved* ratio: RECOMP’s summary length is nearly budget-independent, saturating at achieved ratios of 0.05–0.08 on ASQA and ≈ 0.015 on QASPER across all four nominal budgets. All primary comparisons use the same successful questions across all 21 settings: 944 on ASQA and 976 from 271 papers on QASPER. Only identity has failures: 4 budget-check failures on ASQA, and 15 budget-check plus 14 prompt-overflow failures on QASPER; their question IDs are excluded from every primary setting. Full-cohort descriptive aggregates remain available in the evaluation artifacts.

QASPER context and token counts. The context contains the supplied paper’s abstract and all full-text paragraphs, sentence-segmented for compression; no top- k retrieval truncation is applied. On the 976 common questions, full-source token counts have mean 4,672, median 4,557, and range 897–11,418. At nominal 0.25, RECOMP’s mean compressed context is 60.3 tokens and its mean complete generator input is 125.4 tokens. The reported achieved fraction, 0.01525, is the mean of per-question context/full-source ratios, not the ratio of these token means. At nominal 0.50, this gap between nominal and achieved fractions is about 32.8-fold.

4.2 Statistical Analysis and Metric Audit

Common cohorts and estimands. Primary summaries and comparisons use the intersection of successful question IDs across identity and all twenty compressed settings. Each question has equal weight in citation precision, citation recall, answer quality, and evidence F1; claims and citations are not pooled across questions. Undefined evidence or recovery metrics remain undefined rather than becoming zeros. Verification-rate means exclude answers with zero claims; paired verification comparisons exclude a question if ei-

ther member has zero claims. The analysis artifacts report those exclusions and retain full-cohort descriptive means separately.

Inference. We use pointwise 95% percentile bootstrap intervals for each paired mean difference (Dror et al., 2018), resampling 944 questions on ASQA and 271 papers on QASPER with replacement. All retained questions from a sampled QASPER paper travel together, and each replicate divides its total difference by the number of sampled questions, preserving the question-weighted estimand. Every QASPER contrast retains all 271 papers after metric-specific exclusions. Two-sided randomization tests swap paired labels within questions on ASQA and jointly within papers on QASPER. These tests assume label exchangeability within a resampling unit under the null. Both procedures use 10,000 replicates and seed 13, with independent random-number streams for bootstrap sampling and randomization. Monte Carlo p -values use $(b + 1)/(10,000 + 1)$, where b counts randomizations at least as extreme as the observed absolute difference. The initial plan specified 134 core contrasts and 10 recovery-threshold contrasts before inspecting the revised inferential results. After that analysis and a cache-coverage audit, we added four recovered-source versus summary-verification contrasts at nominal 0.25 for RECOMP and LLMingua-2 on both datasets. The extension plan was recorded before running these four tests, and Holm adjustment was recomputed over all 148 raw p -values. We denote Holm-adjusted values by p_H . All reported adjusted values use this expanded family; original effects, intervals, and raw p -values are unchanged. This is a post hoc camera-ready analysis, not a preregistration. Intervals are pointwise, not simultaneous; statistical-support statements use $p_H < 0.05$. Failure to reject a difference does not establish equivalence or preservation. All inference is conditional on the fixed generator outputs, recovery procedure, and evaluator.

Precision-code audit. The manuscript’s precision definition requires the concatenated cited evidence to entail a claim before any individual citation receives credit. A code audit found that the previous implementation could credit an individually supporting citation even when joint entailment failed. A saved ASQA answer reproduced this discrepancy, and a regression test now checks the required gate through the evaluation pipeline.

Compressor	$\hat{\rho}$	P_g	P_e	U_s	U_m	U_r
<i>ASQA</i>						
Identity	1.00	0.51	0.51	0.44	0.44	0.44
Rerank top- k	0.20	0.69	0.69	0.36	0.37	0.36
Extractive	0.26	0.49	0.49	0.46	0.46	0.46
ECR	0.26	0.48	0.49	0.46	0.45	0.46
RECOMP	0.08	0.12	0.86	1.00	0.17	0.88
LLMLingua-2	0.26	0.10	0.53	1.00	0.51	0.90
<i>QASPER</i>						
Identity	1.00	0.48	0.48	0.53	0.53	0.53
Rerank top- k	0.25	0.39	0.39	0.59	0.60	0.59
Extractive	0.25	0.43	0.43	0.48	0.48	0.48
ECR	0.25	0.46	0.46	0.45	0.45	0.45
RECOMP	0.02	0.06	0.55	1.00	0.50	0.95
LLMLingua-2	0.25	0.11	0.42	1.00	0.65	0.91

Table 1: Citation precision and unsupported rates at nominal budget 0.25 (identity: full context), on 944 ASQA and 976 QASPER questions. Notation follows Section 3; no-provenance declared-span rates are structural. Two zero-claim RECOMP answers are excluded from QASPER verification rates.

All saved answers are rescored with the corrected definition using the same DeBERTa checkpoint in float32; no answers are regenerated. The audit also computes legacy precision from the same new NLI scores to isolate the gate’s effect from numerical inference differences and cohort selection. On the common ASQA cohort, the gate reduces identity precision from 0.5622 to 0.5086, affecting 98 of 944 questions. On QASPER it reduces identity precision from 0.4992 to 0.4826, affecting 61 of 976 questions. These are deterministic scorer-rule comparisons using fixed NLI scores, not new model-performance tests. Input, source-code, checkpoint, and tokenizer hashes accompany persistent per-question records and resumable score caches. These primary scores remain dependent on this single evaluator; the correction does not constitute independent validation. The 512-token NLI input limit can truncate concatenated evidence, so the joint gate is also conditional on that bounded input.

Appendix B reports all 148 tested differences, including those not retained as claims; structural provenance rejections are descriptive.

5 Results

Emitted and grounded scores diverge (H1). Figures 1 and 2 show the frontier. On ASQA at nominal budget 0.25 (achieved ratio 0.08), RECOMP receives *emitted* citation precision of 0.86 against its summaries and *grounded* preci-

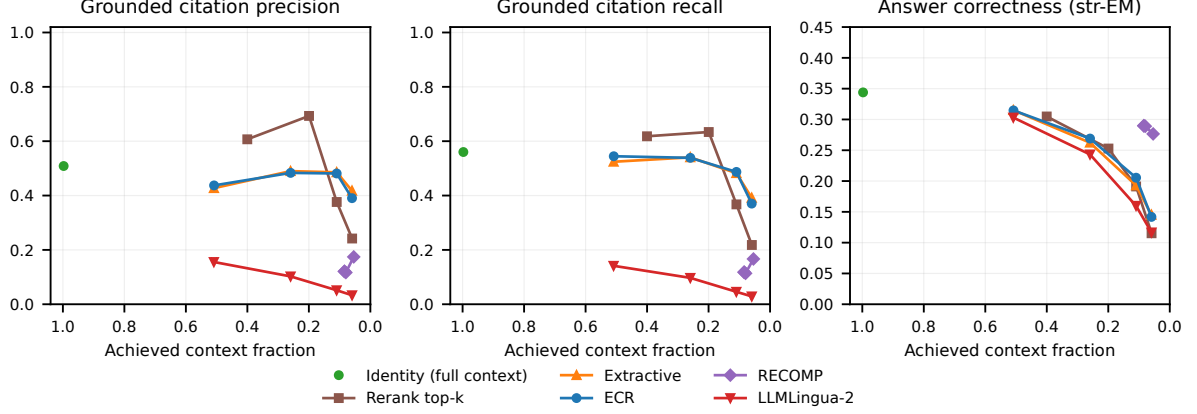


Figure 1: The attribution-compression frontier on 944 common ASQA questions. Compression increases rightward (x : achieved fraction of retrieved context). Curves show descriptive means under the shared evaluator and recovery protocol; paired difference intervals and tests appear in Appendix B. The right panel reports str-EM.

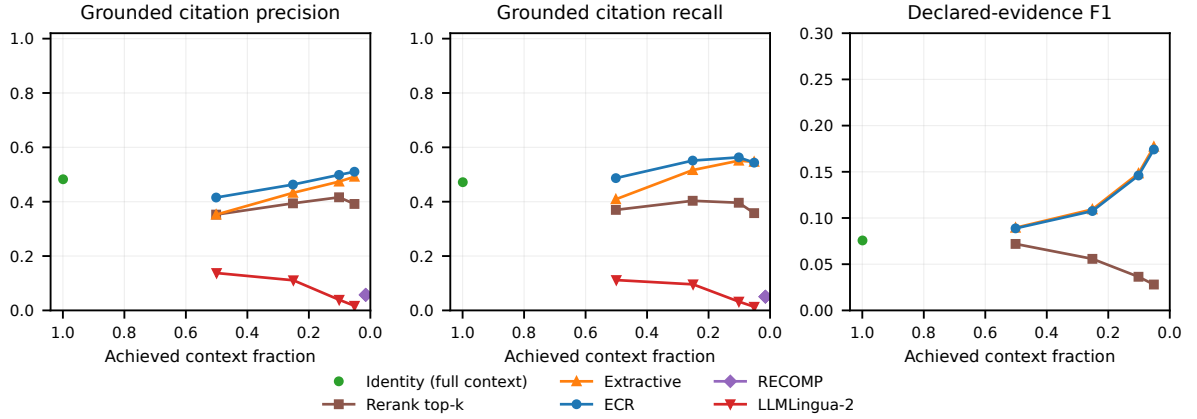


Figure 2: The frontier on 976 common QASPER questions from 271 papers, axes as in Figure 1. The third panel scores each compressor’s *declared* evidence against gold paragraphs; it is undefined for no-provenance methods. Small plotted differences do not establish equivalence.

sion of 0.12 against source spans; its mean perquestion recovery rate is 0.41. The paired precision gap is 0.740 (95% CI [0.711, 0.767], $p_H = 0.0148$). The gap combines effects of compression, bounded recovery, and NLI judgments; these measurements do not separate their contributions. On QASPER the corresponding scores are 0.55 and 0.06 (achieved ratio 0.015), with gap 0.491 [0.457, 0.525] and $p_H = 0.0148$. LLMingua-2’s grounded precision means decrease across budgets: 0.16, 0.10, 0.05, 0.03 on ASQA and 0.14, 0.11, 0.04, 0.02 on QASPER. All no-provenance settings score below their dataset’s identity reference (0.51 on ASQA, 0.48 on QASPER), with $p_H = 0.0148$ for each comparison.

Extraction and reranking trade off citation scores and answer quality. ASQA extraction yields grounded precision 0.43, 0.49, 0.49, and

0.42 across achieved ratios 0.51, 0.26, 0.11, and 0.06. Relative to identity, precision is lower at nominal 0.50 and 0.05 ($p_H = 0.0148$); the intermediate differences are unresolved ($p_H = 1$), not evidence of preservation. Its str-EM decreases from identity’s 0.34 through 0.32, 0.26, and 0.19 to 0.15, with every adjacent decrease supported ($p_H = 0.0148$). ASQA reranking at nominal 0.25 increases precision from 0.51 to 0.69 ($\Delta = 0.184$, CI [0.149, 0.219], $p_H = 0.0148$), while recall rises from 0.56 to 0.63 and str-EM falls from 0.34 to 0.25 (both $p_H = 0.0148$). The corresponding QASPER precision difference is negative ($\Delta = -0.089$, CI [-0.124, -0.052], $p_H = 0.0148$). These dataset-specific results do not isolate a causal effect of citation granularity.

Declared provenance determines verification coverage (H4). Table 1 contrasts declared-span

and summary checking. For RECOMP on ASQA the unsupported rates are 1.00 and 0.17, respectively; the corresponding QASPER values are 1.00 and 0.50. LLMingua-2 has rates of 1.00 vs. 0.51 on ASQA and 1.00 vs. 0.65 on QASPER. Every 1.00 here is structural: no declared source evidence reaches the span verifier, so claims fail without an entailment judgment. These values do not establish that all claims lack recoverable source support. Recovered-source verification now supplies a direct claim-level comparison: for RECOMP, unsupported rates are 0.88 on ASQA and 0.95 on QASPER. Their paired differences from summary checking are 0.712 (CI [0.683, 0.739]) and 0.450 [0.417, 0.483], both $p_H = 0.0148$. LLMingua-2’s recovered-source rates are 0.90 and 0.91, with differences 0.394 [0.362, 0.426] and 0.253 [0.225, 0.282] (both $p_H = 0.0148$). These gaps persist when recovery is attempted, but remain conditional on the bounded search and shared NLI evaluator. For provenance-declaring methods, displayed rate differences are below 0.01 in magnitude; no mode comparison survives Holm correction, which does not establish equivalence.

ECR’s advantage is limited to one tested setting. On QASPER, ECR improves grounded precision over extraction at nominal 0.50 ($\Delta = 0.063$, CI [0.033, 0.092], $p_H = 0.0148$). Its other QASPER precision differences and all four ASQA differences lack adjusted statistical support. The tightest QASPER ECR setting also does not establish an advantage over identity ($\Delta = 0.028$, CI [−0.010, 0.064], $p_H = 1$). ECR’s QASPER declared-evidence F1 is slightly lower than extraction at every budget, by 0.00065–0.00295 (all $p_H < 0.05$). Thus the results support a setting-specific precision gain, not general superiority or lossless redundancy removal.

6 Analysis

Re-attribution varies across no-provenance methods. The re-attribution rate measures whether at least one candidate source snippet passes the NLI entailment threshold for a unit without declared provenance. On ASQA, RECOMP’s rates are 0.41, 0.41, 0.42, and 0.39 across nominal budgets 0.5 through 0.05, excluding examples without summary units. Averaged over these examples, roughly three-fifths of its units have no passing candidate; this does not establish that the source contains no supporting

Compressor	Gen. in	Ver. span	Ver. sum.	Total
Identity	798	371	382	1169
Rerank top- k	210	171	174	381
Extractive	277	112	114	389
ECR	277	118	107	395
RECOMP	127	0	84	127
LLMingua-2	253	0	240	253

Table 2: Mean input tokens on 944 ASQA questions at nominal budget 0.25 (identity: full context). Total includes generator and declared-span verification input; compression, output, and recovery are excluded.

evidence. LLMingua-2’s ASQA rate means decrease from 0.56 to 0.40, 0.25, and 0.20 as budgets tighten; the first two adjacent decreases have $p_H = 0.0148$, while the final decrease is unresolved ($p_H = 0.242$). Recovery coverage depends on method, budget, and dataset as well as the search and evaluator; these observations provide no universal ceiling on recoverable support.

Verification adds input tokens. Table 2 reports a generator-input-plus-verifier-input proxy, measured in generator-tokenizer tokens before NLI truncation. It excludes compressor inference, generated output, recovery, and differences in model-specific tokenization or per-token cost. The declared-span column is zero for no-provenance compressors because this policy rejects their claims without an NLI call. That zero does not describe the cost of recovering and checking source evidence.

Evidence selection under pressure. The declared-evidence panel of Figure 2 shows increasing F1 for extraction and ECR as budgets tighten, from about 0.09 to 0.18 and 0.17, respectively. Each adjacent increase is supported ($p_H = 0.0148$). Reranking’s F1 means decrease from 0.07 to 0.06, 0.04, and 0.03; the first two decreases have $p_H = 0.0148$, but the final difference is unresolved ($p_H = 1$). These are evidence-F1 results, not evidence-precision estimates or a controlled test of selection granularity.

Sensitivity to the re-attribution threshold. The re-attributability protocol depends on the entailment threshold τ . Table 3 sweeps τ from 0.3 to 0.8 for both no-provenance methods on ASQA (citation scoring fixed at the production threshold of 0.5). Grounded precision ranges from 0.15 to 0.08 for RECOMP and from 0.16 to 0.04 for

τ	RECOMP		LLMLingua-2	
	P_g	reattr.	P_g	reattr.
0.3	0.15	0.48 ⁿ⁼⁹²⁹	0.16	0.54
0.4	0.13	0.44 ⁿ⁼⁹²⁹	0.13	0.47
0.5	0.12	0.41 ⁿ⁼⁹²⁹	0.10	0.40
0.6	0.11	0.38 ⁿ⁼⁹²⁹	0.08	0.34
0.7	0.09	0.34 ⁿ⁼⁹²⁹	0.05	0.27
0.8	0.08	0.31 ⁿ⁼⁹²⁹	0.04	0.20

Table 3: Recovery-threshold sensitivity on 944 ASQA questions at nominal budget 0.25; citation scoring stays at $\tau = 0.5$. P_g : grounded precision; reattr.: re-attribution rate. Superscripts count defined values; dashes mark undefined metrics.

LLMLingua-2, while recovery rates fall from 0.48 to 0.31 and from 0.54 to 0.20, respectively. Emitted precision exceeds grounded precision at every tested recovery threshold for both methods ($p_H = 0.0148$ throughout). This is sensitivity analysis of one operating choice, not independent validation: the evaluator, top-10 lexical search, and single-snippet support requirement remain fixed.

Second-evaluator sensitivity. We rescore 100 questions per dataset with TRUE T5-XXL (Honovich et al., 2022), covering full-context identity plus extraction, RECOMP, and LLMLingua-2 at nominal 0.25. With source mappings fixed, RECOMP’s emitted/grounded precision is 0.78/0.13 on ASQA and 0.44/0.01 on QASPER. The paired gaps are 0.65 [0.55, 0.75] and 0.43 [0.33, 0.53], respectively. When TRUE also selects recovered spans, grounded precision is 0.14 and 0.03. All 16 planned emitted-minus-grounded precision and recall gaps across both methods, datasets, and recovery modes are positive ($p_H = 0.0024$ in a separate 24-test family). None of the eight cross-evaluator grounded-precision differences passes Holm adjustment; this does not establish equivalence or validate the primary numbers. Appendix C reports all comparisons, the 4,096-token input configuration, decoder anomalies, and documented restarts. The pattern persists under a second scoring family, with recovery and calibration limitations retained.

7 Discussion

What laundering looks like. Consider an ASQA example under RECOMP at nominal budget 0.25. Asked “How many state parks are there in Virginia?”, the generator answers “There are 34 state parks in Virginia [1]”, while Document [1] states “Virginia has 34 state parks and 17 state

forests...”. The citation receives full emitted precision, but Document [1] is the compressor’s summary with no declared source link. In another example, the answer “The man who killed [them] is not mentioned in the retrieved documents [1]” cites a summary containing that same assertion of absence. Both illustrate consistency with intermediate text without a declared path to source evidence; neither example alone establishes an independently verified hallucination.

Implications for practice. Five operational recommendations follow from this evaluation setup. (1) *Verify against source spans*: consistency with compressor output does not establish support from locatable evidence. Report whether spans are declared or recovered and how support is judged. (2) *Report achieved compression*: nominal fractions overstate achieved context size by up to 32.8-fold here (Section 4.1). (3) *Include verification in cost reporting*: report generator and verifier input lengths together, while identifying which other costs remain unmeasured (Table 2). Zero declared-span cost can mean that no evidence reached the verifier. (4) *Evaluate post-hoc recovery explicitly*: missed recovery can reflect search errors or missing support (Section 6). (5) *Document citable-unit granularity*: comparisons differ in their source units as well as their datasets. Controlled ablations are needed to isolate the effect of unit size.

What dominating the frontier would require. Future compressors should merge redundant evidence while retaining validated source links and accounting for verification cost. ECR’s limited measured gains motivate independent calibration and replication before claiming general improvements.

Failure modes and excluded examples. All 33 failures occur in identity runs: 4 budget-check failures on ASQA, and 15 budget-check plus 14 prompt-overflow failures on QASPER. No compressed run fails on these examples. Primary comparisons exclude the same failed question IDs from every setting, so their estimand is performance conditional on identity success.

8 Conclusion

We measured citation attribution across context-compression methods and budgets under a fixed generator and primary NLI evaluator. Emitted and grounded scores diverge, but their magnitudes

depend on recovery and evaluation choices that have not been independently calibrated. Declared-span verification rejects claims lacking declared evidence by policy; it does not determine whether support could be recovered. Paired comparisons quantify sampling uncertainty conditional on the fixed outputs and evaluator, with only limited evidence for ECR’s advantage. We recommend reporting grounded alongside emitted citation scores and achieved alongside nominal compression. The TRUE subsample retains the emitted–grounded gaps; human calibration and broader replication remain necessary.

9 Limitations

Evaluator dependence. Primary scores use DeBERTa-v3-base, trained on MNLI, FEVER, and ANLI, at $\tau = 0.5$, without human calibration. Using the same model to select recovered spans and score support can reinforce systematic errors; the resulting scores are evaluator-dependent estimates, not validated rates of source support or factual correctness. This dependence also limits conclusions about method rankings and the size of emitted–grounded gaps. The 200-question TRUE audit checks a second scoring family, but fixed recovery retains DeBERTa selection and TRUE recovery reuses one judge. Different input windows and nonbinary decoder outputs further limit numerical comparison. Human source-level calibration remains necessary (Rashkin et al., 2023).

Single generator. All results use Qwen2.5-7B-Instruct with greedy decoding. Frontier shapes may differ across model families and scales; a second-family replication is planned.

Dataset coverage. We evaluate two English datasets (ASQA and QASPER) with distinct retrieval granularities, but this does not cover multilingual settings, open-web retrieval, or longer-context regimes.

Re-attribution hyperparameters. The top-10 lexical candidate pool can miss semantically matching snippets, and requiring one snippet to entail a unit can miss support distributed across snippets. Consequently, a failed recovery does not establish that the source contains no support. Table 3 varies the recovery threshold while holding the scoring threshold fixed; it does not validate the evaluator or test candidate-pool size, retrieval strategy, or multi-snippet recovery.

Verification scope. Declared-span verification rejects claims lacking declared evidence by construction, even if support could be recovered. Its 1.00 unsupported rate for no-provenance runs is a provenance-availability diagnostic, not a calibrated hallucination rate. Recovered-source verification is reported at nominal 0.25 for the two no-provenance methods and remains dependent on the same bounded recovery procedure and evaluator.

Residual error rows. Only identity fails on 4 ASQA and 29 QASPER questions; every primary setting is evaluated on the resulting common successful subset. This pairing prevents comparisons from using different questions, but conditions results on identity success and omits those difficult cases. Sampling intervals condition on the fixed generated outputs and evaluator; they do not include systematic NLI error or variation across generation runs.

Nominal versus achieved budgets. Abstractive compressors do not reliably hit nominal token budgets; RECOMP often produces much less context than requested. We report *achieved* compression ratios, but comparisons at a shared nominal budget do not establish performance at matched achieved context sizes.

Reproducibility Statement

Saved generation outputs, input hashes, and pinned evaluator revisions accompany per-question rescoring records. The fixed contrast plans and paired-analysis reports regenerate the numerical tables and frontier figures without further model calls. The harness, configurations, protocol, and per-question records will be publicly released upon publication.

References

- Bernd Bohnet, Vinh Q. Tran, Pat Verga, Roei Aharoni, Daniel Andor, Livio Baldini Soares, Massimiliano Ciaramita, Jacob Eisenstein, Kuzman Ganchev, Jonathan Herzig, Kai Hui, Tom Kwiatkowski, Ji Ma, Jianmo Ni, Lierni Sestorain Saralegui, Tal Schuster, William W. Cohen, Michael Collins, Dipanjan Das, and 3 others. 2022. [Attributed question answering: Evaluation and modeling for attributed large language models](#). *Preprint*, arXiv:2212.08037.
- Samuel R. Bowman, Gabor Angeli, Christopher Potts, and Christopher D. Manning. 2015. [A large annotated corpus for learning natural language inference](#). In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, pages 632–642, Lisbon, Portugal. Association for Computational Linguistics.
- Tong Chen, Hongwei Wang, Sihao Chen, Wenhao Yu, Kaixin Ma, Xinran Zhao, Hongming Zhang, and Dong Yu. 2024. [Dense X retrieval: What retrieval granularity should we use?](#) In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 15159–15177, Miami, Florida, USA. Association for Computational Linguistics.
- Xin Cheng, Xun Wang, Xingxing Zhang, Tao Ge, Si-Qing Chen, Furu Wei, Huishuai Zhang, and Dongyan Zhao. 2024. [xRAG: Extreme context compression for retrieval-augmented generation with one token](#). In *Advances in Neural Information Processing Systems*, volume 37, pages 109487–109516.
- Nadezhda Chirkova, Thibault Formal, Vassilina Nikoulina, and Stéphane Clinchant. 2025. [Provence: Efficient and robust context pruning for retrieval-augmented generation](#). In *International Conference on Learning Representations*.
- Pradeep Dasigi, Kyle Lo, Iz Beltagy, Arman Cohan, Noah A. Smith, and Matt Gardner. 2021. [A dataset of information-seeking questions and answers anchored in research papers](#). In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4599–4610, Online. Association for Computational Linguistics.
- Rotem Dror, Gili Baumer, Segev Shlomov, and Roi Reichart. 2018. [The hitchhiker’s guide to testing statistical significance in natural language processing](#). In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1383–1392, Melbourne, Australia. Association for Computational Linguistics.
- Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, Dasha Metropolitan, Robert Osazuwa Ness, and Jonathan Larson. 2024. [From local to global: A graph RAG approach to query-focused summarization](#). *Preprint*, arXiv:2404.16130.
- Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. 2023. [Enabling large language models to generate text with citations](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 6465–6488, Singapore. Association for Computational Linguistics.
- Tao Ge, Jing Hu, Lei Wang, Xun Wang, Si-Qing Chen, and Furu Wei. 2024. [In-context autoencoder for context compression in a large language model](#). In *International Conference on Learning Representations*.
- Shuo Guan. 2026. [Attributable by construction: Claim-anchored provenance for multi-document summarization](#). *Preprint*, arXiv:2606.23989.
- Or Honovich, Roei Aharoni, Jonathan Herzig, Hagai Taitelbaum, Vered Cohen, Doron Kukliansky, Thomas Scialom, Idan Szpektor, Avinatan Hassidim, and Yossi Matias. 2022. [TRUE: Re-evaluating factual consistency evaluation](#). In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 3905–3920, Seattle, United States. Association for Computational Linguistics.
- Nabil Ibtihaz and Daisuke Kihara. 2026. [Protein-STORY: Semantic text-oriented representation yields biologically meaningful protein embeddings](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 883–897, San Diego, California, United States. Association for Computational Linguistics.
- Sarthak Jain and Byron C. Wallace. 2019. [Attention is not explanation](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 3543–3556, Minneapolis, Minnesota. Association for Computational Linguistics.
- Yeonseok Jeong, Jinsu Kim, Dohyeon Lee, and Seungwon Hwang. 2025. [ECoRAG: Evidentiality-guided compression for long context RAG](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 26607–26628, Vienna, Austria. Association for Computational Linguistics.
- Huiqiang Jiang, Qianhui Wu, Xufang Luo, Dongsheng Li, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. 2024. [LongLLMLingua: Accelerating and enhancing LLMs in long context scenarios via prompt compression](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1658–1677, Bangkok, Thailand. Association for Computational Linguistics.
- Dongwon Jung, Qin Liu, Tenghao Huang, Ben Zhou, and Muhao Chen. 2025. [Familiarity-aware evidence compression for retrieval-augmented generation](#). In *Findings of the Association for Compu-*

- tational Linguistics: EMNLP 2025*, pages 16181–16196, Suzhou, China. Association for Computational Linguistics.
- Tushar Khot, Ashish Sabharwal, and Peter Clark. 2018. [SciTail: A textual entailment dataset from science question answering](#). In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, pages 5189–5197. Association for the Advancement of Artificial Intelligence.
- Weronika Łajewska, Momchil Hardalov, Laura Aina, Neha Anna John, Hang Su, and Lluís Marquez. 2025. [Understanding and improving information preservation in prompt compression for LLMs](#). In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 17520–17541, Suzhou, China. Association for Computational Linguistics.
- Aiyu Li, Qian Peng, and Bin Chen. 2026a. [Efficiency vs. verifiability in evidence-aware RAG: Does prompt compression preserve citation grounding?](#) In *Proceedings of the Second Workshop on Customizable NLP: Progress and Challenges in Customizing NLP for a Domain, Application, Group, or Individual (CustomNLP4U)*, pages 202–215, San Diego, California, USA. Association for Computational Linguistics.
- Yilong Li, Suman Banerjee, and Tong Che. 2026b. [EMBER: Efficient memory via budgeted evidence retention for long-horizon agents](#). *Preprint*, arXiv:2606.05894.
- Francesco Locatello, Dirk Weissenborn, Thomas Unterthiner, Aravindh Mahendran, Georg Heigold, Jakob Uszkoreit, Alexey Dosovitskiy, and Thomas Kipf. 2020. [Object-centric learning with slot attention](#). In *Advances in Neural Information Processing Systems*, volume 33, pages 11525–11538.
- Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. [FACTScore: Fine-grained atomic evaluation of factual precision in long form text generation](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12076–12100, Singapore. Association for Computational Linguistics.
- Jesse Mu, Xiang Lisa Li, and Noah Goodman. 2023. [Learning to compress prompts with gist tokens](#). In *Advances in Neural Information Processing Systems*, volume 36, pages 19327–19352.
- Zhuoshi Pan, Qianhui Wu, Huiqiang Jiang, Menglin Xia, Xufang Luo, Jue Zhang, Qingwei Lin, Victor Rühle, Yuqing Yang, Chin-Yew Lin, H. Vicky Zhao, Lili Qiu, and Dongmei Zhang. 2024. [LLMLingua-2: Data distillation for efficient and faithful task-agnostic prompt compression](#). In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 963–981, Bangkok, Thailand. Association for Computational Linguistics.
- Hannah Rashkin, Vitaly Nikolaev, Matthew Lamm, Lora Aroyo, Michael Collins, Dipanjan Das, Slav Petrov, Gaurav Singh Tomar, Iulia Turc, and David Reitter. 2023. [Measuring attribution in natural language generation models](#). *Computational Linguistics*, 49(4):777–840.
- David Rau, Shuai Wang, Hervé Déjean, Stéphane Clinchant, and Jaap Kamps. 2025. [Context embeddings for efficient answer generation in retrieval-augmented generation](#). In *Proceedings of the Eighteenth ACM International Conference on Web Search and Data Mining*, pages 493–502. Association for Computing Machinery.
- Parth Sarthi, Salman Abdullah, Aditi Tuli, Shubh Khanna, Anna Goldie, and Christopher D. Manning. 2024. [RAPTOR: Recursive abstractive processing for tree-organized retrieval](#). In *International Conference on Learning Representations*.
- Tal Schuster, Adam Fisch, and Regina Barzilay. 2021. [Get your vitamin C! robust fact verification with contrastive evidence](#). In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 624–643, Online. Association for Computational Linguistics.
- Xue Wen Tan and Stanley Kok. 2025. [SMARTe: Slot-based method for accountable relational triple extraction](#). *Preprint*, arXiv:2504.12816.
- James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. 2018. [FEVER: a large-scale dataset for fact extraction and VERification](#). In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 809–819, New Orleans, Louisiana. Association for Computational Linguistics.
- Jonas Wallat, Maria Heuss, Maarten de Rijke, and Avishek Anand. 2025. [Correctness is not faithfulness in retrieval augmented generation attributions](#). In *Proceedings of the 2025 International ACM SIGIR Conference on Innovative Concepts and Theories in Information Retrieval, ICTIR '25*, pages 22–32. Association for Computing Machinery.
- Sarah Wiegreffe and Yuval Pinter. 2019. [Attention is not not explanation](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 11–20, Hong Kong, China. Association for Computational Linguistics.
- Adina Williams, Nikita Nangia, and Samuel R. Bowman. 2018. [A broad-coverage challenge corpus for sentence understanding through inference](#). In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*,

Volume 1 (Long Papers), pages 1112–1122, New Orleans, Louisiana. Association for Computational Linguistics.

Theodora Worledge, Tatsunori Hashimoto, and Carlos Guestrin. 2024. [The extractive-abstractive spectrum: Uncovering verifiability trade-offs in LLM generations](#). *Preprint*, arXiv:2411.17375.

Fangyuan Xu, Weijia Shi, and Eunsol Choi. 2024. [RECOMP: Improving retrieval-augmented LMs with compression and selective augmentation](#). In *International Conference on Learning Representations*.

Jiajie Zhang, Yushi Bai, Xin Lv, Wanjun Gu, Danning Liu, Minhao Zou, Shulin Cao, Lei Hou, Yuxiao Dong, Ling Feng, and Juanzi Li. 2025. [LongCite: Enabling LLMs to generate fine-grained citations in long-context QA](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 5098–5122, Vienna, Austria. Association for Computational Linguistics.

Yuan Zhang, Jason Baldridge, and Luheng He. 2019. [PAWS: Paraphrase adversaries from word scrambling](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 1298–1308, Minneapolis, Minnesota. Association for Computational Linguistics.

Qiming Zhu, Shunian Chen, Rui Yu, Zhehao Wu, and Benyou Wang. 2026. [From lossy to verified: A provenance-aware tiered memory for agents](#). *Preprint*, arXiv:2602.17913.

Appendices

A Results on Common Questions

Tables 4 and 5 report all primary settings. ρ : nominal budget; $\hat{\rho}$: achieved ratio; P_g/R_g : grounded citation precision/recall; P_e : emitted citation precision; reattr.: re-attribution rate (units without declared provenance recovering at least one source span under the NLI protocol); U_s/U_m : unsupported-claim rate under declared-span-only and summary-checked verification; str-EM: answer string exact match (ASQA); EF1: declared-evidence F1 (QASPER only). Every setting uses the same 944 ASQA or 976 QASPER successful questions; n denotes that common cohort, with metric-specific omissions identified in the tables. Scores are question-weighted means over defined values; superscript $n = \dots$ gives the non-missing count when smaller than the cohort, and dashes mark undefined metrics. QASPER verification rates exclude 1, 2, 3, and 2 zero-claim RECOMP answers at nominal budgets 0.50, 0.25, 0.10, and 0.05, respectively; other settings use the full cohort. The 1.00 declared-span rates for no-provenance methods are structural; this verification mode does not attempt recovery. Exploratory slot-compressor runs are excluded. RECOMP emits identical per-question contexts at nominal budgets 0.50 and 0.25 on ASQA, and at 0.50, 0.25, and 0.10 on QASPER. Separate generation passes are retained as distinct operating points.

B Complete Paired Comparisons

Section 4.2 describes the common cohorts, estimands, inference, and precision-code correction. Tables 6–10 report all 148 planned contrasts, including those not retained as claims; EF1 effects and intervals use four decimal places, and other effects and intervals use three.

Table notation. A/Q denote ASQA/QASPER; Id, RR, X, RC, and LL denote identity, reranking, extraction, RECOMP, and LLMingua-2. Setting labels give compressor/nominal budget, optionally followed by the recovery threshold. P/R denote precision/recall, with e/g for emitted/grounded; EM is str-EM, EF1 is evidence F1, and Rec is re-attribution rate. $U_s/U_m/U_r$ denote declared-span, summary, and recovered-source unsupported rates. Δ is the paired left-minus-right mean difference; a single endpoint applies to both settings, and a pair lists left-minus-right metrics. CI is a pointwise 95% bootstrap interval; p_H is the two-sided randomization p -value after Holm adjustment across all 148 comparisons; n counts included questions.

Recovered-source verification. This extension reads existing NLI scores from the sealed cache and makes no new model calls. It uses the same recovered source IDs as citation scoring, but the claim verifier removes repeated source IDs before pooling evidence. Consequently, its unsupported rate need not equal one minus grounded citation recall, which retains repeated cited evidence. Both summary and recovered-source rates exclude the same zero-claim answers: two QASPER RECOMP answers at nominal 0.25; the other three comparisons use their full common cohorts. Input hashes, the extension plan, cache lookups, per-question records, and all four comparisons are retained separately from the original analysis.

Compressor	ρ	$\hat{\rho}$	P_g	R_g	P_e	reattr.	str-EM	U_s	U_m	n
Identity	1	1.00	0.51	0.56	0.51	–	0.34	0.44	0.44	944
Rerank top- k	0.5	0.40	0.61	0.62	0.61	–	0.30	0.38	0.38	944
Rerank top- k	0.25	0.20	0.69	0.63	0.69	–	0.25	0.36	0.37	944
Rerank top- k	0.1	0.11	0.38	0.37	0.38	–	0.19	0.63	0.63	944
Rerank top- k	0.05	0.06	0.24	0.22	0.24	–	0.12	0.78	0.78	944
Extractive	0.5	0.51	0.43	0.52	0.43	–	0.32	0.47	0.48	944
Extractive	0.25	0.26	0.49	0.54	0.49	–	0.26	0.46	0.46	944
Extractive	0.1	0.11	0.49	0.48	0.49	–	0.19	0.52	0.52	944
Extractive	0.05	0.06	0.42	0.39	0.42	–	0.15	0.61	0.61	944
ECR	0.5	0.51	0.44	0.54	0.45	–	0.31	0.45	0.44	944
ECR	0.25	0.26	0.48	0.54	0.49	–	0.27	0.46	0.45	944
ECR	0.1	0.11	0.48	0.49	0.49	–	0.21	0.51	0.51	944
ECR	0.05	0.06	0.39	0.37	0.39	–	0.14	0.63	0.63	944
RECOMP	0.5	0.08	0.12	0.12	0.86	$0.41^{n=929}$	0.29	1.00	0.17	944
RECOMP	0.25	0.08	0.12	0.12	0.86	$0.41^{n=929}$	0.29	1.00	0.17	944
RECOMP	0.1	0.08	0.12	0.11	0.86	$0.42^{n=929}$	0.29	1.00	0.17	944
RECOMP	0.05	0.05	0.17	0.17	0.80	$0.39^{n=928}$	0.28	1.00	0.24	944
LLMLingua-2	0.5	0.51	0.16	0.14	0.58	0.56	0.30	1.00	0.45	944
LLMLingua-2	0.25	0.26	0.10	0.10	0.53	0.40	0.24	1.00	0.51	944
LLMLingua-2	0.1	0.11	0.05	0.05	0.31	0.25	0.16	1.00	0.73	944
LLMLingua-2	0.05	0.06	0.03	0.03	0.16	0.20	0.12	1.00	0.86	944

Table 4: All 21 primary settings on 944 common ASQA questions; question-weighted means. Notation and exclusions follow Appendix A. No-provenance declared-span rates are structural.

Compressor	ρ	$\hat{\rho}$	P_g	R_g	P_e	reattr.	EF1	U_s	U_m	n
Identity	1	1.00	0.48	0.47	0.48	–	0.08	0.53	0.53	976
Rerank top- k	0.5	0.50	0.35	0.37	0.35	–	0.07	0.63	0.63	976
Rerank top- k	0.25	0.25	0.39	0.40	0.39	–	0.06	0.59	0.60	976
Rerank top- k	0.1	0.10	0.42	0.40	0.42	–	0.04	0.60	0.60	976
Rerank top- k	0.05	0.05	0.39	0.36	0.39	–	0.03	0.64	0.64	976
Extractive	0.5	0.50	0.35	0.41	0.35	–	0.09	0.59	0.59	976
Extractive	0.25	0.25	0.43	0.52	0.43	–	0.11	0.48	0.48	976
Extractive	0.1	0.10	0.47	0.55	0.47	–	0.15	0.44	0.45	976
Extractive	0.05	0.05	0.49	0.55	0.49	–	0.18	0.45	0.45	976
ECR	0.5	0.50	0.42	0.49	0.42	–	0.09	0.51	0.51	976
ECR	0.25	0.25	0.46	0.55	0.46	–	0.11	0.45	0.45	976
ECR	0.1	0.10	0.50	0.56	0.50	–	0.15	0.43	0.44	976
ECR	0.05	0.05	0.51	0.54	0.51	–	0.17	0.45	0.46	976
RECOMP	0.5	0.02	0.06	0.05	0.56	$0.44^{n=845}$	–	$1.00^{n=975}$	$0.49^{n=975}$	976
RECOMP	0.25	0.02	0.06	0.05	0.55	$0.44^{n=845}$	–	$1.00^{n=974}$	$0.50^{n=974}$	976
RECOMP	0.1	0.02	0.06	0.05	0.55	$0.44^{n=845}$	–	$1.00^{n=973}$	$0.50^{n=973}$	976
RECOMP	0.05	0.02	0.06	0.05	0.55	$0.44^{n=845}$	–	$1.00^{n=974}$	$0.50^{n=974}$	976
LLMLingua-2	0.5	0.50	0.14	0.11	0.50	0.92	–	1.00	0.61	976
LLMLingua-2	0.25	0.25	0.11	0.10	0.42	0.79	–	1.00	0.65	976
LLMLingua-2	0.1	0.10	0.04	0.03	0.43	0.35	–	1.00	0.63	976
LLMLingua-2	0.05	0.05	0.02	0.01	0.38	0.18	–	1.00	0.67	976

Table 5: All 21 primary settings on 976 common QASPER questions; question-weighted means. Notation and exclusions follow Appendix A. No-provenance declared-span rates are structural.

ID	Data	Left	Right	Endpoint	Δ	95% CI	p_H	n
C001	A	RC/.5	RC/.5	$P_e - P_g$	+0.744	[+0.715, +0.771]	0.0148	944
C002	A	RC/.25	RC/.25	$P_e - P_g$	+0.740	[+0.711, +0.767]	0.0148	944
C003	A	RC/.1	RC/.1	$P_e - P_g$	+0.743	[+0.715, +0.770]	0.0148	944
C004	A	RC/.05	RC/.05	$P_e - P_g$	+0.621	[+0.588, +0.654]	0.0148	944
C005	A	LL/.5	LL/.5	$P_e - P_g$	+0.426	[+0.394, +0.458]	0.0148	944
C006	A	LL/.25	LL/.25	$P_e - P_g$	+0.425	[+0.391, +0.458]	0.0148	944
C007	A	LL/.1	LL/.1	$P_e - P_g$	+0.260	[+0.229, +0.291]	0.0148	944
C008	A	LL/.05	LL/.05	$P_e - P_g$	+0.132	[+0.108, +0.156]	0.0148	944
C009	Q	RC/.5	RC/.5	$P_e - P_g$	+0.499	[+0.464, +0.534]	0.0148	976
C010	Q	RC/.25	RC/.25	$P_e - P_g$	+0.491	[+0.457, +0.525]	0.0148	976
C011	Q	RC/.1	RC/.1	$P_e - P_g$	+0.492	[+0.457, +0.526]	0.0148	976
C012	Q	RC/.05	RC/.05	$P_e - P_g$	+0.496	[+0.462, +0.531]	0.0148	976
C013	Q	LL/.5	LL/.5	$P_e - P_g$	+0.360	[+0.319, +0.402]	0.0148	976
C014	Q	LL/.25	LL/.25	$P_e - P_g$	+0.309	[+0.278, +0.341]	0.0148	976
C015	Q	LL/.1	LL/.1	$P_e - P_g$	+0.388	[+0.354, +0.421]	0.0148	976
C016	Q	LL/.05	LL/.05	$P_e - P_g$	+0.361	[+0.329, +0.395]	0.0148	976
C017	A	RC/.25	RC/.25	$R_e - R_g$	+0.712	[+0.684, +0.740]	0.0148	944
C018	A	RR/.5	Id	P_g	+0.098	[+0.066, +0.131]	0.0148	944
C019	A	RR/.25	Id	P_g	+0.184	[+0.149, +0.219]	0.0148	944
C020	A	RR/.1	Id	P_g	-0.132	[-0.170, -0.095]	0.0148	944
C021	A	RR/.05	Id	P_g	-0.267	[-0.303, -0.230]	0.0148	944
C022	A	X/.5	Id	P_g	-0.081	[-0.116, -0.047]	0.0148	944
C023	A	X/.25	Id	P_g	-0.019	[-0.053, +0.016]	1.0000	944
C024	A	X/.1	Id	P_g	-0.023	[-0.061, +0.014]	1.0000	944
C025	A	X/.05	Id	P_g	-0.091	[-0.129, -0.053]	0.0148	944
C026	A	ECR/.5	Id	P_g	-0.071	[-0.105, -0.038]	0.0148	944
C027	A	ECR/.25	Id	P_g	-0.025	[-0.061, +0.010]	1.0000	944
C028	A	ECR/.1	Id	P_g	-0.027	[-0.064, +0.010]	1.0000	944
C029	A	ECR/.05	Id	P_g	-0.118	[-0.155, -0.080]	0.0148	944
C030	A	RC/.5	Id	P_g	-0.388	[-0.421, -0.354]	0.0148	944

Table 6: Planned comparisons 1–30 of 148. Δ : left-minus-right mean difference; p_H : Holm-adjusted p -value. Setting and metric labels follow Appendix B.

ID	Data	Left	Right	Endpoint	Δ	95% CI	p_H	n
C031	A	RC/.25	Id	P_g	-0.387	[-0.421, -0.354]	0.0148	944
C032	A	RC/.1	Id	P_g	-0.392	[-0.424, -0.359]	0.0148	944
C033	A	RC/.05	Id	P_g	-0.334	[-0.370, -0.299]	0.0148	944
C034	A	LL/.5	Id	P_g	-0.353	[-0.386, -0.320]	0.0148	944
C035	A	LL/.25	Id	P_g	-0.406	[-0.438, -0.374]	0.0148	944
C036	A	LL/.1	Id	P_g	-0.457	[-0.488, -0.427]	0.0148	944
C037	A	LL/.05	Id	P_g	-0.475	[-0.504, -0.447]	0.0148	944
C038	Q	RR/.5	Id	P_g	-0.130	[-0.165, -0.095]	0.0148	976
C039	Q	RR/.25	Id	P_g	-0.089	[-0.124, -0.052]	0.0148	976
C040	Q	RR/.1	Id	P_g	-0.066	[-0.104, -0.029]	0.0280	976
C041	Q	RR/.05	Id	P_g	-0.091	[-0.128, -0.053]	0.0148	976
C042	Q	X/.5	Id	P_g	-0.130	[-0.159, -0.101]	0.0148	976
C043	Q	X/.25	Id	P_g	-0.050	[-0.082, -0.019]	0.0832	976
C044	Q	X/.1	Id	P_g	-0.009	[-0.041, +0.024]	1.0000	976
C045	Q	X/.05	Id	P_g	+0.009	[-0.027, +0.046]	1.0000	976
C046	Q	ECR/.5	Id	P_g	-0.067	[-0.099, -0.035]	0.0148	976
C047	Q	ECR/.25	Id	P_g	-0.020	[-0.050, +0.011]	1.0000	976
C048	Q	ECR/.1	Id	P_g	+0.016	[-0.019, +0.051]	1.0000	976
C049	Q	ECR/.05	Id	P_g	+0.028	[-0.010, +0.064]	1.0000	976
C050	Q	RC/.5	Id	P_g	-0.426	[-0.455, -0.397]	0.0148	976
C051	Q	RC/.25	Id	P_g	-0.425	[-0.455, -0.396]	0.0148	976
C052	Q	RC/.1	Id	P_g	-0.426	[-0.456, -0.397]	0.0148	976
C053	Q	RC/.05	Id	P_g	-0.425	[-0.454, -0.396]	0.0148	976
C054	Q	LL/.5	Id	P_g	-0.345	[-0.381, -0.309]	0.0148	976
C055	Q	LL/.25	Id	P_g	-0.372	[-0.404, -0.338]	0.0148	976
C056	Q	LL/.1	Id	P_g	-0.444	[-0.471, -0.416]	0.0148	976
C057	Q	LL/.05	Id	P_g	-0.466	[-0.493, -0.439]	0.0148	976
C058	A	RR/.25	Id	R_g	+0.074	[+0.039, +0.107]	0.0148	944
C059	A	ECR/.5	X/.5	P_g	+0.010	[-0.018, +0.038]	1.0000	944
C060	A	ECR/.25	X/.25	P_g	-0.006	[-0.036, +0.022]	1.0000	944

Table 7: Planned comparisons 31–60 of 148. Δ : left-minus-right mean difference; p_H : Holm-adjusted p -value. Setting and metric labels follow Appendix B.

ID	Data	Left	Right	Endpoint	Δ	95% CI	p_H	n
C061	A	ECR/.1	X/.1	P_g	-0.004	[-0.029, +0.021]	1.0000	944
C062	A	ECR/.05	X/.05	P_g	-0.027	[-0.049, -0.005]	0.4158	944
C063	Q	ECR/.5	X/.5	P_g	+0.063	[+0.033, +0.092]	0.0148	976
C064	Q	ECR/.25	X/.25	P_g	+0.030	[+0.002, +0.059]	1.0000	976
C065	Q	ECR/.1	X/.1	P_g	+0.025	[-0.001, +0.051]	1.0000	976
C066	Q	ECR/.05	X/.05	P_g	+0.018	[-0.009, +0.046]	1.0000	976
C067	A	LL/.25	LL/.5	P_g	-0.053	[-0.076, -0.029]	0.0148	944
C068	A	LL/.1	LL/.25	P_g	-0.051	[-0.073, -0.030]	0.0148	944
C069	A	LL/.05	LL/.1	P_g	-0.018	[-0.036, -0.001]	1.0000	944
C070	Q	LL/.25	LL/.5	P_g	-0.027	[-0.058, +0.004]	1.0000	976
C071	Q	LL/.1	LL/.25	P_g	-0.072	[-0.096, -0.048]	0.0148	976
C072	Q	LL/.05	LL/.1	P_g	-0.022	[-0.041, -0.005]	0.3836	976
C073	A	RR/.25	RR/.5	P_g	+0.086	[+0.053, +0.117]	0.0148	944
C074	A	RR/.25	RR/.1	P_g	+0.317	[+0.280, +0.354]	0.0148	944
C075	A	RR/.25	RR/.05	P_g	+0.451	[+0.415, +0.487]	0.0148	944
C076	A	RR/.5	Id	EM	-0.039	[-0.055, -0.023]	0.0148	944
C077	A	RR/.25	Id	EM	-0.091	[-0.108, -0.074]	0.0148	944
C078	A	RR/.1	Id	EM	-0.153	[-0.172, -0.134]	0.0148	944
C079	A	RR/.05	Id	EM	-0.229	[-0.249, -0.209]	0.0148	944
C080	A	X/.5	Id	EM	-0.029	[-0.044, -0.014]	0.0148	944
C081	A	X/.25	Id	EM	-0.082	[-0.100, -0.064]	0.0148	944
C082	A	X/.1	Id	EM	-0.151	[-0.170, -0.132]	0.0148	944
C083	A	X/.05	Id	EM	-0.199	[-0.219, -0.179]	0.0148	944
C084	A	ECR/.5	Id	EM	-0.029	[-0.045, -0.014]	0.0216	944
C085	A	ECR/.25	Id	EM	-0.075	[-0.093, -0.057]	0.0148	944
C086	A	ECR/.1	Id	EM	-0.138	[-0.157, -0.120]	0.0148	944
C087	A	ECR/.05	Id	EM	-0.202	[-0.222, -0.183]	0.0148	944
C088	A	RC/.5	Id	EM	-0.054	[-0.070, -0.039]	0.0148	944
C089	A	RC/.25	Id	EM	-0.054	[-0.069, -0.039]	0.0148	944
C090	A	RC/.1	Id	EM	-0.055	[-0.070, -0.040]	0.0148	944

Table 8: Planned comparisons 61–90 of 148. Δ : left-minus-right mean difference; p_H : Holm-adjusted p -value. Setting and metric labels follow Appendix B.

ID	Data	Left	Right	Endpoint	Δ	95% CI	p_H	n
C091	A	RC/.05	Id	EM	-0.067	[-0.084, -0.051]	0.0148	944
C092	A	LL/.5	Id	EM	-0.041	[-0.055, -0.026]	0.0148	944
C093	A	LL/.25	Id	EM	-0.101	[-0.118, -0.083]	0.0148	944
C094	A	LL/.1	Id	EM	-0.184	[-0.203, -0.165]	0.0148	944
C095	A	LL/.05	Id	EM	-0.227	[-0.248, -0.207]	0.0148	944
C096	A	RR/.25	RR/.5	EM	-0.052	[-0.064, -0.040]	0.0148	944
C097	A	RR/.1	RR/.25	EM	-0.062	[-0.077, -0.046]	0.0148	944
C098	A	RR/.05	RR/.1	EM	-0.076	[-0.092, -0.060]	0.0148	944
C099	A	X/.25	X/.5	EM	-0.053	[-0.068, -0.037]	0.0148	944
C100	A	X/.1	X/.25	EM	-0.069	[-0.084, -0.055]	0.0148	944
C101	A	X/.05	X/.1	EM	-0.048	[-0.062, -0.034]	0.0148	944
C102	A	ECR/.25	ECR/.5	EM	-0.046	[-0.060, -0.032]	0.0148	944
C103	A	ECR/.1	ECR/.25	EM	-0.064	[-0.079, -0.048]	0.0148	944
C104	A	ECR/.05	ECR/.1	EM	-0.064	[-0.079, -0.049]	0.0148	944
C105	A	RC/.25	RC/.5	EM	+0.000	[-0.002, +0.002]	1.0000	944
C106	A	RC/.1	RC/.25	EM	-0.001	[-0.006, +0.004]	1.0000	944
C107	A	RC/.05	RC/.1	EM	-0.012	[-0.022, -0.003]	0.3030	944
C108	A	LL/.25	LL/.5	EM	-0.060	[-0.076, -0.045]	0.0148	944
C109	A	LL/.1	LL/.25	EM	-0.083	[-0.100, -0.067]	0.0148	944
C110	A	LL/.05	LL/.1	EM	-0.043	[-0.057, -0.030]	0.0148	944
C111	Q	ECR/.5	X/.5	EF1	-0.0007	[-0.0011, -0.0002]	0.0374	976
C112	Q	ECR/.25	X/.25	EF1	-0.0018	[-0.0025, -0.0012]	0.0148	976
C113	Q	ECR/.1	X/.1	EF1	-0.0022	[-0.0035, -0.0010]	0.0148	976
C114	Q	ECR/.05	X/.05	EF1	-0.0030	[-0.0046, -0.0012]	0.0185	976
C115	Q	X/.25	X/.5	EF1	+0.0199	[+0.0170, +0.0227]	0.0148	976
C116	Q	X/.1	X/.25	EF1	+0.0389	[+0.0343, +0.0435]	0.0148	976
C117	Q	X/.05	X/.1	EF1	+0.0289	[+0.0231, +0.0346]	0.0148	976
C118	Q	ECR/.25	ECR/.5	EF1	+0.0187	[+0.0160, +0.0214]	0.0148	976
C119	Q	ECR/.1	ECR/.25	EF1	+0.0385	[+0.0339, +0.0430]	0.0148	976
C120	Q	ECR/.05	ECR/.1	EF1	+0.0281	[+0.0222, +0.0340]	0.0148	976

Table 9: Planned comparisons 91–120 of 148. Δ : left-minus-right mean difference; p_H : Holm-adjusted p -value. Setting and metric labels follow Appendix B.

ID	Data	Left	Right	Endpoint	Δ	95% CI	p_H	n
C121	Q	RR/.25	RR/.5	EF1	-0.0162	[-0.0225, -0.0097]	0.0148	976
C122	Q	RR/.1	RR/.25	EF1	-0.0194	[-0.0275, -0.0112]	0.0148	976
C123	Q	RR/.05	RR/.1	EF1	-0.0084	[-0.0171, +0.0008]	1.0000	976
C124	A	LL/.25	LL/.5	Rec	-0.156	[-0.194, -0.117]	0.0148	944
C125	A	LL/.1	LL/.25	Rec	-0.153	[-0.192, -0.114]	0.0148	944
C126	A	LL/.05	LL/.1	Rec	-0.050	[-0.086, -0.015]	0.2418	944
C127	A	Id	Id	$U_s - U_m$	-0.002	[-0.007, +0.002]	1.0000	944
C128	A	RR/.25	RR/.25	$U_s - U_m$	-0.003	[-0.008, +0.002]	1.0000	944
C129	A	X/.25	X/.25	$U_s - U_m$	+0.000	[-0.002, +0.001]	1.0000	944
C130	A	ECR/.25	ECR/.25	$U_s - U_m$	+0.009	[+0.002, +0.016]	0.3045	944
C131	Q	Id	Id	$U_s - U_m$	-0.001	[-0.005, +0.003]	1.0000	976
C132	Q	RR/.25	RR/.25	$U_s - U_m$	-0.002	[-0.006, +0.001]	1.0000	976
C133	Q	X/.25	X/.25	$U_s - U_m$	-0.008	[-0.014, -0.003]	0.0627	976
C134	Q	ECR/.25	ECR/.25	$U_s - U_m$	-0.001	[-0.007, +0.004]	1.0000	976
C135	A	RC/.25	RC/.25/.3	$P_e - P_g$	+0.714	[+0.685, +0.743]	0.0148	944
C136	A	RC/.25	RC/.25/.4	$P_e - P_g$	+0.730	[+0.701, +0.758]	0.0148	944
C137	A	RC/.25	RC/.25/.6	$P_e - P_g$	+0.753	[+0.726, +0.780]	0.0148	944
C138	A	RC/.25	RC/.25/.7	$P_e - P_g$	+0.766	[+0.739, +0.792]	0.0148	944
C139	A	RC/.25	RC/.25/.8	$P_e - P_g$	+0.779	[+0.752, +0.805]	0.0148	944
C140	A	LL/.25	LL/.25/.3	$P_e - P_g$	+0.372	[+0.338, +0.406]	0.0148	944
C141	A	LL/.25	LL/.25/.4	$P_e - P_g$	+0.400	[+0.367, +0.433]	0.0148	944
C142	A	LL/.25	LL/.25/.6	$P_e - P_g$	+0.449	[+0.415, +0.481]	0.0148	944
C143	A	LL/.25	LL/.25/.7	$P_e - P_g$	+0.474	[+0.441, +0.506]	0.0148	944
C144	A	LL/.25	LL/.25/.8	$P_e - P_g$	+0.488	[+0.455, +0.520]	0.0148	944
C145	A	RC/.25	RC/.25	$U_r - U_m$	+0.712	[+0.683, +0.739]	0.0148	944
C146	A	LL/.25	LL/.25	$U_r - U_m$	+0.394	[+0.362, +0.426]	0.0148	944
C147	Q	RC/.25	RC/.25	$U_r - U_m$	+0.450	[+0.417, +0.483]	0.0148	974
C148	Q	LL/.25	LL/.25	$U_r - U_m$	+0.253	[+0.225, +0.282]	0.0148	976

Table 10: Planned comparisons 121–148 of 148. Δ : left-minus-right mean difference; p_H : Holm-adjusted p -value. Setting and metric labels follow Appendix B.

C Second-Evaluator Sensitivity Audit

We sample 100 questions per dataset from sorted common-cohort IDs using a fresh Python seed-13 random generator; QASPER spans 84 papers. Full-context identity and nominal-0.25 extraction, RECOMP, and LLMLingua-2 contribute 800 unchanged saved answers, with 1,627 claims and 1,890 citation markers. The prior DeBERTa values are the original scores for these same sampled answers.

Model and decoding. We use google/t5_xxl_true_nli_mixture (Honovich et al., 2022), revision aa6cfe1dd4, with full revision and checkpoint hashes in the manifest. This released mixture uses SNLI, MNLI, FEVER, SciTail, PAWS, and VitaminC (Bowman et al., 2015; Williams et al., 2018; Thorne et al., 2018; Khot et al., 2018; Zhang et al., 2019; Schuster et al., 2021), rather than the original TRUE paper’s ANLI recipe. MNLI and FEVER overlap DeBERTa’s training mixture. Inference uses Transformers 4.57.6, the slow T5 tokenizer, BF16, batch size four, and one NVIDIA B200. The prompt places compressed text or source evidence after premise: and the unit or claim after hypothesis:.

We use an AutoAIS-style exact-one decision (Bohnet et al., 2022; Gao et al., 2023): generate at most ten tokens deterministically, strip surrounding whitespace, and count only 1 as entailment. Every other string scores zero; these are binary decisions, not calibrated probabilities. Unlike our rule, reference AutoAIS/ALCE code compares decoded strings without additional whitespace normalization. We retain normalized nonbinary strings, but not every raw output, so normalization’s effect cannot be quantified from the cache.

Evidence mappings. *Fixed recovery* preserves declared provenance and DeBERTa-recovered source IDs while TRUE scores support. *TRUE recovery* reevaluates all top-10 lexical candidates for undeclared units, including DeBERTa-rejected candidates, then scores with TRUE. Both retain the corrected precision gate. Fixed recovery retains selection dependence; TRUE recovery reuses one judge. Neither removes bounded-search errors or supplies human calibration.

Recorded amendments. A strict-decoding pilot stopped on a nonbinary output; a second stopped

on an overlong hypothesis. An input-only scan found five undeclared units exceeding 2,048 tokens (maximum hypothesis plus prompt: 2,988), motivating a uniform 4,096-token cap that preserves complete hypotheses and truncates only premises. A later batch-eight run exhausted GPU memory. All interrupted attempts and plans are preserved but excluded; the final batch-four run recomputes every input in a fresh cache. These amendments preceded inspection of any aggregate TRUE comparison report and left the sample, thresholds, and 24 comparisons unchanged. The cap exceeds original TRUE’s 2,048 and DeBERTa’s 512 tokens, so cross-evaluator differences do not isolate model family.

Inference and diagnostics. The separate 24-test family uses pointwise 95% bootstrap intervals and two-sided randomization tests, each with 10,000 replicates and seed 13, question resampling on ASQA and paper clustering on QASPER, with question-weighted means and Holm adjustment. The original 148-test analysis is unchanged. Tables 11–12 report precision means and all planned contrasts. All 16 emitted-grounded gaps remain positive after adjustment; the eight cross-evaluator precision differences remain unresolved, which does not establish equivalence. ECR and other budgets are outside this audit.

The run scored 6,015 unique pairs in 419 seconds of model inference (430 seconds including bookkeeping; setup and excluded attempts are additional). Four inputs required truncation, with maximum pre-truncation length 47,440 tokens. After normalization, 122 pairs (2.03%) produced nonbinary strings, including 103 empty strings; assigning zero does not establish reliable non-entailment. These are pair counts, not fractions of questions affected; the full histogram accompanies the cache. Recovery can disagree substantially: QASPER LLMLingua-2 has 50.3% agreement over 1,000 candidate decisions and mean recovery rates of 0.84 for DeBERTa versus 0.38 for TRUE. These descriptive results support sensitivity analysis, not accuracy against human judgments.

Data	Method	Prior DeBERTa		TRUE fixed		TRUE recovery	
		P_e	P_g	P_e	P_g	P_e	P_g
A	Identity	0.467	0.467	0.563	0.563	0.563	0.563
A	Extractive	0.402	0.402	0.497	0.497	0.497	0.497
A	RECOMP	0.790	0.090	0.780	0.130	0.780	0.140
A	LLMLingua-2	0.480	0.070	0.415	0.023	0.415	0.010
Q	Identity	0.487	0.487	0.498	0.498	0.498	0.498
Q	Extractive	0.461	0.461	0.391	0.391	0.391	0.391
Q	RECOMP	0.550	0.020	0.440	0.010	0.440	0.030
Q	LLMLingua-2	0.490	0.118	0.387	0.076	0.387	0.019

Table 11: Precision on the same sampled saved answers: 100 ASQA (A) and 100 QASPER (Q) questions (84 QASPER papers). P_e/P_g : emitted/grounded precision. Mappings and evaluation details follow Appendix C; superscripts give nonmissing counts and dashes undefined values.

ID	Data	Method	View	Endpoint	Δ	95% CI	p_H	n	k
T01	A	RC	F	$P_e - P_g$	+0.650	[+0.550, +0.750]	0.0024	100	100
T02	A	RC	F	$R_e - R_g$	+0.640	[+0.540, +0.735]	0.0024	100	100
T03	A	LL	F	$P_e - P_g$	+0.392	[+0.290, +0.495]	0.0024	100	100
T04	A	LL	F	$R_e - R_g$	+0.367	[+0.273, +0.463]	0.0024	100	100
T05	Q	RC	F	$P_e - P_g$	+0.430	[+0.326, +0.530]	0.0024	100	84
T06	Q	RC	F	$R_e - R_g$	+0.425	[+0.320, +0.525]	0.0024	100	84
T07	Q	LL	F	$P_e - P_g$	+0.311	[+0.234, +0.390]	0.0024	100	84
T08	Q	LL	F	$R_e - R_g$	+0.233	[+0.176, +0.293]	0.0024	100	84
T09	A	RC	T	$P_e - P_g$	+0.640	[+0.540, +0.740]	0.0024	100	100
T10	A	RC	T	$R_e - R_g$	+0.637	[+0.540, +0.730]	0.0024	100	100
T11	A	LL	T	$P_e - P_g$	+0.405	[+0.312, +0.502]	0.0024	100	100
T12	A	LL	T	$R_e - R_g$	+0.375	[+0.287, +0.468]	0.0024	100	100
T13	Q	RC	T	$P_e - P_g$	+0.410	[+0.309, +0.505]	0.0024	100	84
T14	Q	RC	T	$R_e - R_g$	+0.405	[+0.303, +0.505]	0.0024	100	84
T15	Q	LL	T	$P_e - P_g$	+0.368	[+0.286, +0.452]	0.0024	100	84
T16	Q	LL	T	$R_e - R_g$	+0.263	[+0.204, +0.325]	0.0024	100	84
T17	A	Id	$F - D$	P_g	+0.096	[+0.011, +0.185]	0.2376	100	100
T18	A	X	$F - D$	P_g	+0.095	[+0.024, +0.168]	0.0880	100	100
T19	A	RC	$F - D$	P_g	+0.040	[−0.010, +0.100]	0.8810	100	100
T20	A	LL	$F - D$	P_g	−0.047	[−0.097, +0.000]	0.6299	100	100
T21	Q	Id	$F - D$	P_g	+0.010	[−0.074, +0.098]	0.9887	100	84
T22	Q	X	$F - D$	P_g	−0.070	[−0.134, −0.008]	0.1827	100	84
T23	Q	RC	$F - D$	P_g	−0.010	[−0.025, +0.000]	0.9887	100	84
T24	Q	LL	$F - D$	P_g	−0.042	[−0.095, +0.011]	0.6299	100	84

Table 12: All 24 planned TRUE robustness contrasts, separate from the original 148-test analysis. $D/F/T$: prior DeBERTa/TRUE fixed/TRUE recovery; n/k : questions/clusters. Δ : left-minus-right mean difference; p_H : Holm-adjusted within this 24-test family. 10,000 replicates, seed 13. Other notation follows Appendix B.